

Extremal Betti Corners Form an Antichain

Source-faithful proof draft for conjecture 00000000508

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The source states that the extremal positions of $\beta_{k,k+d}$ for a monomial ideal form an antichain. We use the standard Betti-diagram coordinates (k, d) : k is homological degree and d is total degree minus homological degree. The order is coordinatewise. Signed $d \in \mathbb{Z}$ is allowed throughout.

Let $\beta : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{N}$ be any numerical table. No finiteness, resolution, or support hypothesis is imposed. We use the definition in Bayer–Charalambous–Popescu [1]: an entry at the raw indices (i, j) is extremal when $\beta_{i,j} \neq 0$ and

$$\beta_{l,r} = 0 \quad \text{for every } l \geq i, \quad r \geq j + 1, \quad r - l \geq j - i.$$

Write $\text{Ext}(\beta)$ for the set of shifted positions (k, d) at which the raw entry $\beta_{k,k+d}$ is extremal, and put

$$P_\beta = \{(k, d) \in \mathbb{N} \times \mathbb{Z} : \beta_{k,k+d} \neq 0\}.$$

Lemma 1. *For every numerical table β and every $p = (k, d)$, $p \in \text{Ext}(\beta)$ if and only if p is a maximal element of P_β in the coordinatewise order.*

Proof. Suppose that p is extremal and $q = (l, u) \in P_\beta$ satisfies $p \leq q$. Thus $l \geq k$ and $u \geq d$. If $q \neq p$, at least one of these inequalities is strict. Since both coordinates are integral, $l + u \geq k + d + 1$. Taking $r = l + u$, the extremality conditions hold: $l \geq k$, $r \geq (k + d) + 1$, and $r - l = u \geq d$. They force $\beta_{l,l+u} = 0$, contradicting $q \in P_\beta$. Therefore $q = p$, so p is maximal.

Conversely, suppose that p is maximal in P_β . In particular, $\beta_{k,k+d} \neq 0$. Let l, r satisfy the three vanishing-region conditions for the raw index $(k, k + d)$. If $\beta_{l,r} \neq 0$, then $q = (l, r - l) \in P_\beta$ and $p \leq q$. Maximality gives $q \leq p$, so $q = p$. This implies $r = k + d$, contradicting $r \geq k + d + 1$. Consequently $\beta_{l,r} = 0$, as required. $\square$

Theorem 2. *For every numerical table $\beta : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{N}$, $\text{Ext}(\beta)$ is an antichain in $\mathbb{N} \times \mathbb{Z}$ with the coordinatewise order.*

Proof. By Lemma 1, its elements are precisely the maximal elements of P_β . If two such elements p, q satisfy $p \leq q$, maximality of p gives $q \leq p$, hence $p = q$. This excludes comparability of distinct elements. $\square$

Corollary 3. *If (k, d) and (l, u) are extremal positions and $k < l$, then $u < d$.*

Proof. If instead $d \leq u$, then $(k, d) \leq (l, u)$, contradicting Theorem 2 because their homological degrees differ. $\square$

The graded Betti numbers of any monomial ideal give a numerical table of the type in Theorem 2. Substitution therefore gives the source’s antichain assertion with no additional algebraic premise. The Lean proof deliberately establishes this universal numerical fact; it does not construct a particular ideal or assume that its extremal support is an antichain.

Coordinate clarification. The conclusion concerns (k, d) for $\beta_{k, k+d}$. It does not assert the same product-order claim for raw indices (k, j) . For example, the numerical table with only $\beta_{0,2} = \beta_{1,2} = 1$ nonzero has two extremal raw entries comparable at $(0, 2) \leq (1, 2)$. Their shifted positions $(0, 2)$ and $(1, 1)$ are incomparable. This example is only a coordinate diagnostic; it is not asserted to be the Betti table of a monomial ideal. The source does not explicitly spell out the order, so acceptance requires independent review of the standard shifted coordinate interpretation.

Formal verification scope. The corresponding Lean declarations are listed individually to keep their names legible:

- `TLMC508.rawExtremal_iff_maximal_shiftedSupport`
- `TLMC508.extremal_positions_antichain`
- `TLMC508.shifted_degree_strictly_decreases`

Local compilation and an axiom audit are worker evidence. Deterministic package checks and source-bound semantic acceptance remain with the manager and independent reviewer. This document is an unaccepted source draft.

References

- [1] D. Bayer, H. Charalambous, and S. Popescu, *Extremal Betti Numbers and Applications to Monomial Ideals*, Manuscript, arXiv:math/9804052. Primary definition in the introduction, p. 1 of the manuscript: https://www.math.columbia.edu/~bayer/papers/Betti_BCP99/Betti_BCP99.pdf.